Appendix B. Simple Connectivity and Jordan Curve Theorem

Equivalent forms of simple connectivity, the Jordan curve theorem, general Cauchy theorem

Contents

1 Equivalent formulations of simple connectivity **2**

2 The Jordan curve theorem **4**

2.1 Proof of a general form of Cauchy's theorem 8

Overview

Recall (Chapter 3) that Ω is *simply connected* if any two curves in Ω with the same end-points are homotopic; then

$$\int_\eta f(\zeta)\,d\zeta = 0 \tag{1}$$

for every closed curve $\eta \subset \Omega$ and every f holomorphic in Ω. In this appendix we prove:

(I) Ω is simply connected if and only if it is *holomorphically simply connected*, i.e. (1) holds for all closed $\eta \subset \Omega$ and all f holomorphic in Ω.

(II) A bounded region is simply connected if and only if its complement is connected.

(III) Ω is simply connected if and only if no curve in Ω winds around a point of Ω^c.

Second, what region does a simple closed curve γ "enclose"? We prove the Jordan curve theorem for piecewise-smooth curves (γ^c has two components, the bounded one simply connected, both with boundary γ) and a general Cauchy theorem extending the toy-contour version of Chapter 2. "Curve" means "piecewise-smooth curve".

1 Equivalent formulations of simple connectivity

Motivation. Simple connectivity, vanishing integrals, winding numbers and connected complements are equivalent tests.

Theorem 1.1

A region Ω is holomorphically simply connected if and only if Ω is simply connected.

Proof. If Ω is simply connected it is holomorphically simply connected (Chapter 3, Corollary 5.3). Conversely, if $\Omega \neq \mathbb{C}$ is holomorphically simply connected, the proof of the Riemann mapping theorem (Theorem 3.1) applies verbatim, since it uses simple connectivity only for a logarithm (and square root) $\log g(z) = c + \int_{z_0}^z g'/g$ of a non-vanishing g, well defined by (1). So Ω is conformally equivalent to $\mathbb{D}$, hence simply connected (conformal maps carry homotopies to homotopies). ∎

Theorem 1.2

If Ω is a bounded region in $\mathbb{C}$, then Ω is simply connected if and only if Ω^c is connected.

Boundedness is needed: a strip is simply connected with disconnected complement. With complements in the Riemann sphere the conclusion holds for all regions.

Proof that Ω^c connected $\Rightarrow$ Ω simply connected. By Theorem 1.1 it suffices to show $\int_\eta f = 0$ for $\eta \subset \Omega$ closed and f holomorphic in Ω. With $d(z, \Omega^c) = \inf\{|z - w| : w \in \Omega^c\}$, let

$$K_\delta = \{z \in \overline{\Omega} : d(z, \Omega^c) \geq \delta\}.$$

K_δ is compact (Ω is bounded), $K_\delta \subset \Omega$, and $\eta \subset K_\delta$ for small δ since η is compact. We show that K_δ^c is connected.

Suppose $K_\delta^c = O_1 \cup O_2$ with O_1, O_2 disjoint, open, non-empty, and set $F_i = O_i \cap \Omega^c$. Since $\Omega^c \subset K_\delta^c$, $\Omega^c = F_1 \cup F_2$ with $F_1 \cap F_2 = \emptyset$.

Step F_1 is closed. Let $z_n \in F_1$, $z_n \to z$. Then $z \in \Omega^c$ (Ω^c is closed), so $z \in O_1 \cup O_2$. If $z \in O_2$, then $z_n \in O_2$ for large n as O_2 is open, contradicting $z_n \in O_1$. So $z \in O_1$, i.e. $z \in F_1$. Likewise F_2 is closed.

Step $F_1 \neq \emptyset$. Otherwise $\Omega^c \subset O_2$. For $w \in O_1$ ($w \notin K_\delta$) there is $z \in \Omega^c$ with $|w - z| < \delta$; the segment $[w, z]$ lies in $K_\delta^c = O_1 \cup O_2$ and meets both, a contradiction. Likewise $F_2 \neq \emptyset$.

So Ω^c would be disconnected; hence K_δ^c is connected, and Runge (Chapter 2, Theorem 5.7) gives polynomials $P_n \to f$ uniformly on $K_\delta \supset \eta$, so $\int_\eta f = \lim \int_\eta P_n = 0$. ■

The converse is proved after Theorem 1.4, using winding numbers.

Winding numbers

If γ is closed and $z \notin \gamma$, each loop of γ around z changes $\frac{1}{2\pi}\arg(\zeta - z)$ by ± 1. Since $\log w = \log|w| + i \arg w$, the quantity $\frac{1}{2\pi i}\int_\gamma \frac{d\zeta}{\zeta - z}$ should count these loops.

Definition (Winding number)

The **winding number** (or **index**) of a closed curve γ around $z \notin \gamma$ is

$$W_\gamma(z) = \frac{1}{2\pi i}\int_\gamma \frac{d\zeta}{\zeta - z}.$$

For $\gamma(t) = e^{ikt}$, $0 \le t \le 2\pi$ ($k \in \mathbb{N}$), $W_\gamma(z) = k$ for $|z| < 1$ and 0 for $|z| > 1$; for e^{-ikt} the values are $-k$ and 0. For a positively oriented toy contour, $W_\gamma = 1$ in its interior and 0 in its exterior.

Lemma 1.3

Let γ be a closed curve in $\mathbb{C}$.

(i) If $z \notin \gamma$, then $W_\gamma(z) \in \mathbb{Z}$.

(ii) If z, w lie in the same connected component of γ^c, then $W_\gamma(z) = W_\gamma(w)$.

(iii) If z lies in the unbounded component of γ^c, then $W_\gamma(z) = 0$.

Proof. (i) Let $G(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s) - z}\, ds$, continuous and, off finitely many points, differentiable with $G' = \gamma'/(\gamma - z)$. Then $H = (\gamma - z)e^{-G}$ has $H' = 0$ off finitely many points, so $H(1) = H(0)$; as $\gamma(1) = \gamma(0) \neq z$, $e^{-G(1)} = 1$ and $G(1) = 2\pi i\, W_\gamma(z) \in 2\pi i\mathbb{Z}$.

(ii) W_γ is a continuous function of $z \in \gamma^c$ (the integrand depends continuously on z, uniformly on γ), integer-valued by (i), hence constant on each connected component.

(iii) $|W_\gamma(z)| \le \frac{\ell(\gamma)}{2\pi\, d(z,\gamma)} \to 0$ as $|z| \to \infty$; since W_γ is constant on the unbounded component by (ii), it is 0 there. ■

Theorem 1.4

A bounded region Ω is simply connected if and only if $W_\gamma(z) = 0$ for every closed curve $\gamma \subset \Omega$ and every $z \notin \Omega$.

Proof. If Ω is simply connected and $z \notin \Omega$, then $f(\zeta) = 1/(\zeta - z)$ is holomorphic in Ω, and Cauchy's theorem gives $W_\gamma(z) = 0$.

Conversely, by the part of Theorem 1.2 already proved, it suffices that Ω^c be connected. Otherwise $\Omega^c = F_1 \cup F_2$, disjoint closed non-empty, and only one is unbounded ($\Omega^c \supset \{|z| > R\}$), say F_2; so F_1 is compact. We find a closed $\gamma \subset \Omega$ and $w \in F_1$ with $W_\gamma(w) \neq 0$, using: ■

Lemma 1.5

Under the above assumptions, let $w \in F_1$. There is a finite collection $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ of closed squares of a uniform grid $\mathcal{G}$ such that:

(i) w is in the interior of Q_1;

(ii) the interiors of Q_j and Q_k are disjoint for $j \neq k$;

(iii) F_1 is contained in the interior of $\bigcup_j Q_j$;

(iv) $\bigcup_j Q_j$ is disjoint from F_2;

(v) the boundary of $\bigcup_j Q_j$ lies in Ω and is a finite union of disjoint simple closed polygonal curves.

Proof of Theorem 1.4, assuming Lemma 1.5. Orient each ∂Q_j positively. Since w is interior to Q_1 and $w \notin Q_j$ for $j > 1$,

$$\sum_{j=1}^{n} \frac{1}{2\pi i} \int_{\partial Q_j} \frac{d\zeta}{\zeta - w} = 1. \tag{2}$$

Shared sides cancel; the remaining sides, with $\bigcup Q_j$ on the left, form the polygons $\gamma_1, \dots, \gamma_M$ of (v), so $\sum_j W_{\gamma_j}(w) = 1$ and some $W_{\gamma_{j_0}}(w) \neq 0$ with $\gamma_{j_0} \subset \Omega$, $w \notin \Omega$: a contradiction. ∎

Proof of Lemma 1.5. $d = d(F_1, F_2) > 0$. Take a grid $\mathcal{G}_0$ of closed squares of side $\delta < d/100$ with w the center of a square R_1, and let $\mathcal{R} = \{R_1, \dots, R_m\}$ be those meeting F_1.

Step (i)–(iv) and $\partial \bigcup R_j \subset \Omega$. (i) and (ii) are clear. Points of $\bigcup R_j$ are within $\sqrt{2}\delta < d$ of F_1: (iv). Every grid square containing $z \in F_1$ is in $\mathcal{R}$, and these cover a neighborhood of z: (iii). A *boundary side* (side of exactly one square of $\mathcal{R}$) is a side of a square outside $\mathcal{R}$, so it misses F_1, and F_2 by (iv). So $\partial \bigcup R_j \subset \Omega$.

Step Bad vertices. End-points of boundary sides are boundary vertices. At a grid vertex P there are 0, 2 or 4 boundary sides, 4 exactly when two diagonally opposite squares are in $\mathcal{R}$ and the other two are not; such P is *bad*.

Step Refinement. Cut each square of $\mathcal{G}_0$ into 9 subsquares (grid $\mathcal{G}$), take the $9m$ subsquares of $\mathcal{R}$, Q_1 the middle one of R_1 (centered at w), and at each bad vertex P add the two subsquares with corner P in the two missing big squares; call the result $\mathcal{Q} = \{Q_1, \dots, Q_n\}$. All four small squares at P are now in $\mathcal{Q}$, so P is interior. Of the other corners of an added subsquare s, the two on sides of $\mathcal{G}_0$ have three surrounding small squares in $\mathcal{Q}$ and the inner one only s; added squares at different bad vertices do not touch. So $\mathcal{Q}$ has no bad vertices. (i)–(iv) persist (added squares are within $3\delta < d$ of F_1), and new boundary sides lie in big squares outside $\mathcal{R}$, missing F_1 and F_2; so $\partial \bigcup Q_j \subset \Omega$.

Step (v). Each boundary vertex now meets exactly two boundary sides, one entering and one leaving (orientation with $\bigcup Q_j$ on the left; check one, two adjacent or three squares). From $[a_1, a_2]$ continue uniquely $[a_2, a_3], \dots$; let $a_m = a_k$, $k < m$, be the first repetition. If $k > 1$, a_k meets three sides $[a_{k-1}, a_k]$, $[a_k, a_{k+1}]$, $[a_{m-1}, a_m]$, impossible; so $a_1 \cdots a_m$ is a simple closed polygon. Removing it and repeating gives finitely many disjoint simple closed polygons (each vertex lies on one). ∎

Proof of Theorem 1.2, converse. If Ω is bounded and simply connected with disconnected complement, the proof of Theorem 1.4 gives a closed $\gamma \subset \Omega$ and $w \notin \Omega$ with $W_\gamma(w) \neq 0$, contradicting its first part. ∎

2 The Jordan curve theorem

Motivation. The Jordan curve theorem guarantees that "the interior of a curve" exists; it yields a general Cauchy theorem.

The curves in the statements are piecewise-smooth, but the proofs also use curves that are merely continuous (the curves γ_η below).

Theorem 2.1 (Jordan arc theorem)

Let γ be a simple (non-closed) piecewise-smooth curve in the plane. Then γ^c is an open connected set whose boundary is precisely γ.

Theorem 2.2 (Jordan curve theorem)

For a simple closed piecewise-smooth γ, γ^c consists of two disjoint connected open sets: the bounded, simply connected **interior** Ω_γ and the unbounded **exterior** $\mathcal{U}_\gamma$. With the appropriate orientation of γ,

$$W_\gamma(z) = \begin{cases} 1 & z \in \Omega_\gamma, \\ 0 & z \in \mathcal{U}_\gamma. \end{cases}$$

Remark

Both theorems hold for merely continuous curves, but the proofs are harder; the piecewise-smooth case suffices for many applications.

Theorem 2.3

If f is holomorphic in the interior Ω_γ of a simple closed curve γ, then $\int_\eta f(\zeta)\,d\zeta = 0$ for every closed curve $\eta \subset \Omega_\gamma$.

Proof [Supplement]. Ω_γ is a simply connected region by Theorem 2.2, so this is Cauchy's theorem, Corollary 5.3 of Chapter 3. ∎

Idea of the proof of Theorem 2.1. As γ^c is open, show it is pathwise connected: join $z, w \notin \gamma$ by a curve; if it meets γ, go from z and w to points z', w' near γ, then travel "parallel" to γ, around an end-point if necessary. On a smooth piece, $i\gamma'(t)$ is normal and $\gamma(t) + i\eta\gamma'(t)$ is a parallel curve.

In the next three lemmas and two propositions, γ_0 is a *simple smooth* curve with arc-length parametrization $\gamma : [0, L] \to \mathbb{C}$, $|\gamma'| = 1$; its *interior points* are the $\gamma(t)$, $0 < t < L$ (unrelated to Theorem 2.2).

Lemma 2.4

For $\eta \in \mathbb{R}$ let γ_η be the continuous curve $\gamma_\eta(t) = \gamma(t) + i\eta\gamma'(t)$, $0 \le t \le L$. There is $\eta_1 > 0$ such that $\gamma_0 \cap \gamma_\eta = \emptyset$ whenever $0 < |\eta| < \eta_1$.

Proof.

Step Local version. For $s, t \in [0, L]$,

$$\gamma_\eta(t) - \gamma(s) = \int_s^t \gamma'(u)\,du + i\eta\gamma'(t) = \int_s^t [\gamma'(u) - \gamma'(t)]\,du + (t - s + i\eta)\gamma'(t).$$

By uniform continuity of γ' there is $\delta > 0$ with $|\gamma'(x) - \gamma'(y)| < 1/2$ for $|x - y| < \delta$. If $|s - t| < \delta$, then using $|\gamma'(t)| = 1$ and $2|a + ib| \ge |a| + |b|$,

$$|\gamma_\eta(t) - \gamma(s)| \ge |t - s + i\eta| - \frac{|t - s|}{2} \ge \frac{|\eta|}{2}.$$

So $\gamma_\eta(t) \neq \gamma(s)$ when $|t-s| < \delta$ and $\eta \neq 0$.

Step Global version. Choose $0 = t_0 < \cdots < t_n = L$ with $t_{k+1} - t_k < \delta/2$, and set $I_k = \{t : |t - t_k| \leq \delta/4\}$, $J_k = \{t : |t - t_k| < \delta/2\}$, $J_k' = \{t \in [0, L] : |t - t_k| \geq \delta/2\}$ (intersected with $[0, L]$). The I_k cover $[0, L]$. If $t \in I_k$ and $s \in J_k$ then $|t - s| < \delta$, so

$$\gamma_\eta(I_k) \cap \gamma(J_k) = \emptyset \qquad (\eta \neq 0). \tag{3}$$

Since γ_0 is simple, the compact sets $\gamma(I_k)$ and $\gamma(J_k')$ are disjoint, at distance $d_k > 0$. For $t \in I_k$, $s \in J_k'$,

$$|\gamma_\eta(t) - \gamma(s)| \geq |\gamma(t) - \gamma(s)| - |\eta| \geq d_k - |\eta| \geq d_k/2 \quad \text{if } |\eta| \leq d_k/2,$$

so

$$\gamma_\eta(I_k) \cap \gamma(J_k') = \emptyset \qquad (|\eta| < d_k/2). \tag{4}$$

With $\eta_1 = \min_k d_k/2$, (3) and (4) give $\gamma_0 \cap \gamma_\eta = \emptyset$ for $0 < |\eta| < \eta_1$. ■

Lemma 2.5

Let $z \notin \gamma_0$ and let $z_0 = \gamma(t_0)$, $0 < t_0 < L$, be a closest point of γ_0 to z (so if z is closer to an interior point than to the end-points). Then $z = \gamma(t_0) + i\eta\gamma'(t_0)$, $\eta \neq 0$ real, i.e. $z \in \gamma_\eta$.

Proof. Near t_0, $z - \gamma(t) = z - \gamma(t_0) - \gamma'(t_0)(t - t_0) + o(|t - t_0|)$, so

$$|z - z_0|^2 \leq |z - \gamma(t)|^2 = |z - z_0|^2 - 2(t - t_0)\,\mathrm{Re}\left[(z - \gamma(t_0))\overline{\gamma'(t_0)}\right] + o(|t - t_0|).$$

Since $t - t_0$ takes both signs, $\mathrm{Re}[(z - \gamma(t_0))\overline{\gamma'(t_0)}] = 0$, i.e. $(z - \gamma(t_0))\overline{\gamma'(t_0)} = i\eta$ with $\eta \in \mathbb{R}$. As $|\gamma'(t_0)| = 1$, $\overline{\gamma'(t_0)} = 1/\gamma'(t_0)$, so $z - \gamma(t_0) = i\eta\gamma'(t_0)$; and $\eta \neq 0$ because $z \notin \gamma_0$. ■

For $z \in \gamma_\eta$, $w \in \gamma_\epsilon$ as in Lemma 2.5, z, w are on the **same side** of γ_0 if η, ϵ have the same sign, on **opposite sides** otherwise. This refers to the chosen representations; we do not claim exclusivity. Same-side points are joined along a parallel curve; opposite sides require going around an end-point.

Proposition 2.6

Let A, B be the end-points of γ_0 and K compact with $\gamma_0 \cap K \subset \{A, B\}$. If $z, w \notin \gamma_0$ are on the same side, each closer to an interior point of γ_0 than to K or to A, B, they can be joined by a continuous curve in $(K \cup \gamma_0)^c$.

Proof. By Lemma 2.5, with $z_0 = \gamma(t_0)$, $w_0 = \gamma(s_0)$ interior points closest to z, w,

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) + i\epsilon_0\gamma'(s_0),$$

with η_0, ϵ_0 of the same sign, say positive, and say $t_0 \leq s_0$.

Step Normal segments. The half-open segment from z to z_0 lies in $(K \cup \gamma_0)^c$: a point $p \neq z_0$ of it in $K \cup \gamma_0$ would have $|z - p| < d(z, K \cup \gamma_0)$. So for small $\eta > 0$, z joins $z_\eta = \gamma(t_0) + i\eta\gamma'(t_0)$, and w joins $w_\eta = \gamma(s_0) + i\eta\gamma'(s_0)$.

Step Parallel curve. Take $\eta < \eta_1$ (Lemma 2.4) and below the (positive) distance from K to $\{\gamma(t) : t_0 \leq t \leq s_0\}$, an arc avoiding A, B. Then $\{\gamma_\eta(t) : t_0 \leq t \leq s_0\}$ joins z_η to w_η, missing γ_0 (Lemma 2.4) and K ($|\gamma_\eta(t) - \gamma(t)| = \eta$). ■

Lemma 2.7

There is $\eta_2 > 0$ such that the half-neighborhood

$$N = \{\gamma(L) + \eta e^{i\theta}\gamma'(L) : -\pi/2 \leq \theta \leq \pi/2,\ 0 < \eta < \eta_2\}$$

is disjoint from γ_0.

Proof. As in Lemma 2.4,

$$\gamma(L)+\eta e^{i\theta}\gamma'(L)-\gamma(t)=\int_t^L[\gamma'(u)-\gamma'(L)]\,du+(L-t+\eta e^{i\theta})\gamma'(L).$$

Choose δ with $|\gamma'(u)-\gamma'(L)|<1/2$ for $|u-L|<\delta$. Since $\cos\theta\geq 0$, $|L-t+\eta e^{i\theta}|^2\geq(L-t)^2+\eta^2\geq\frac{1}{2}(L-t+\eta)^2$. So for $L-\delta<t\leq L$,

$$|\gamma(L)+\eta e^{i\theta}\gamma'(L)-\gamma(t)|\geq\frac{L-t+\eta}{\sqrt{2}}-\frac{L-t}{2}\geq\frac{\eta}{\sqrt{2}}>0.$$

Thus $\gamma(t)\notin N$ for $L-\delta<t\leq L$. Finally, take η_2 less than the distance from $\gamma(L)$ to $\{\gamma(t):0\leq t\leq L-\delta\}$, which is positive since γ_0 is simple. ■

Proposition 2.8

Let A be an end-point of γ_0 and K compact with $\gamma_0\cap K\subset\{A\}$. If $z,w\notin\gamma_0$ are closer to interior points of γ_0 than to K or to the end-points, they can be joined by a continuous curve in $(\gamma_0\cup K)^c$.

Proof. For the same side use Proposition 2.6. For opposite sides, with (after reversal) $A=\gamma(0)$ and $B=\gamma(L)\notin K$, write

$$z=\gamma(t_0)+i\eta_0\gamma'(t_0),\qquad w=\gamma(s_0)-i\epsilon_0\gamma'(s_0),\qquad \eta_0,\epsilon_0>0,$$

and $\tau=\min(t_0,s_0)>0$. Take $\eta>0$ below η_1, η_2 (Lemmas 2.4, 2.7) and the (positive) distance from K to $\{\gamma(t):\tau\leq t\leq L\}$. Join z to $\gamma_\eta(t_0)$ by a normal segment, follow γ_η to $\gamma(L)+i\eta\gamma'(L)$, the half-circle $\gamma(L)+\eta e^{i\theta}\gamma'(L)$, θ from $\pi/2$ to $-\pi/2$, inside N, to $\gamma(L)-i\eta\gamma'(L)$, then $\gamma_{-\eta}$ back to s_0 and the normal segment to w. The parallel pieces and the half-circle miss γ_0 (Lemmas 2.4, 2.7); all pieces are within η of $\{\gamma(t):\tau\leq t\leq L\}$, so they miss K. ■

Proof of Theorem 2.1.

Step Boundary. $O=\gamma^c$ is open, $\partial O\subset\gamma$. Each smooth point $\gamma(t)$ is a limit of points $\gamma(t)+i\eta\gamma'(t)\in O$ (Lemma 2.4 on a small sub-arc; the rest of γ is at positive distance), and smooth points are dense, so $\partial O=\gamma$.

Step Connectedness: induction on the number of smooth pieces. For smooth γ with end-points A,B and $Z,W\notin\gamma$, join Z to W by a polygonal Λ avoiding A,B. If Λ meets γ, its first and last meeting points p, q are interior points. Points $z\in\Lambda$ just before p have $d(z,\gamma)\leq|z-p|\to 0$ while $|z-A|,|z-B|$ stay bounded below; take such $z\notin\gamma$ closer to an interior point than to A,B, and similarly w just after q. Proposition 2.8 with $K=\emptyset$ joins z to w in γ^c.

For n pieces, write $\gamma=K\cup\gamma_0$, K the union of $n-1$ consecutive pieces (compact) and γ_0 smooth, $K\cap\gamma_0=\{A\}$. By induction $Z,W\notin\gamma$ are joined in the open connected K^c by a polygonal curve avoiding the other end-point of γ_0, meeting γ_0 only at interior points. Take z (w) just before the first (after the last) meeting point, closer to an interior point of γ_0 than to K or the end-points; Proposition 2.8 joins them in $(\gamma_0\cup K)^c=\gamma^c$. ■

Proof of Theorem 2.2.

Step At most two components. Fix W outside a large disc containing γ; let $\mathcal{U}$ be the set of points joined to W in γ^c, open, connected, containing the exterior of the disc, and $\Omega=\gamma^c-\mathcal{U}$. Remove a smooth piece γ_0 (end-points A,B), leaving the arc K, $K\cap\gamma_0=\{A,B\}$. For $Z\in\Omega$, Theorem 2.1 for K gives Λ_Z from Z to W missing K; it meets γ_0 (as $Z\notin\mathcal{U}$), at interior points. Choose

$z, w \in \Lambda_Z$ as before, with the pieces $Z \to z$ and $w \to W$ in γ^c. Then z, w are on opposite sides (else Proposition 2.6 joins Z to W). For another $Z_1 \in \Omega$ with z_1, w_1, z and z_1 are on the same side (else z_1 is on w's side and Z_1 joins W), so Proposition 2.6 joins z to z_1 in γ^c, and Z, Z_1 are in one component. So the components are $\mathcal{U}$ and (if non-empty) Ω.

Step Exactly two components. By Lemma 1.3 it suffices to find points with different winding numbers. Fix $z_0 = \gamma(t_0)$ interior to a smooth piece, $v = \gamma'(t_0) \neq 0$, and for $\epsilon > 0$ let

$$z_\epsilon = \gamma(t_0) + i\epsilon v, \qquad w_\epsilon = \gamma(t_0) - i\epsilon v.$$

For small ϵ these avoid γ, and the normal segments $\{\gamma(t_0) \pm isv : 0 < s \leq \epsilon\}$ lie in γ^c (Lemma 2.4, plus the distance from $\gamma(t_0)$ to the rest of γ). By Lemma 1.3(ii),

$$\Delta = |W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)|$$

is a constant integer for all small $\epsilon > 0$. With $a(t) = \gamma(t) - \gamma(t_0)$,

$$\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} = \frac{2i\epsilon v\, \gamma'(t)}{a(t)^2 + \epsilon^2 v^2}.$$

Since $a(t) = v(t - t_0) + o(|t - t_0|)$ and $\gamma'(t) = v + o(1)$,

$$\frac{2i\epsilon v \gamma'(t)}{a(t)^2 + \epsilon^2 v^2} = \frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2}\,(1 + \rho(t, \epsilon)),$$

where for $\sigma > 0$ some $\delta > 0$ gives $|\rho(t, \epsilon)| < \sigma$ for $|t - t_0| \leq \delta$, uniformly in ϵ (as $a^2 + \epsilon^2 v^2 = v^2[(t - t_0)^2 + \epsilon^2] + o((t - t_0)^2)$ and $o((t - t_0)^2)/[(t - t_0)^2 + \epsilon^2] \to 0$). Split $W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)$ at $|t - t_0| = \delta$: on $|t - t_0| \geq \delta$, $\gamma(t)$ stays away from $\gamma(t_0)$ while $z_\epsilon, w_\epsilon \to \gamma(t_0)$, so this part $\to 0$. On the rest,

$$\frac{1}{2\pi i} \int_{|t - t_0| < \delta} \frac{2i\epsilon\, dt}{(t - t_0)^2 + \epsilon^2} = \frac{1}{\pi} \int_{-\delta/\epsilon}^{\delta/\epsilon} \frac{ds}{s^2 + 1} = \frac{2}{\pi} \arctan(\delta/\epsilon) \to 1,$$

and the ρ-term is at most $\frac{\sigma}{\pi} \int_{\mathbb{R}} \frac{ds}{s^2+1} = \sigma$ in absolute value. Letting $\epsilon \to 0$ gives $|\Delta - 1| \leq \sigma$ for every $\sigma > 0$, so $\Delta = 1$.

So z_ϵ, w_ϵ lie in different components, $\mathcal{U}$ (unbounded, $W_\gamma = 0$, Lemma 1.3(iii)) and the bounded Ω, where $W_\gamma = \pm 1$ ($\Delta = 1$), $= 1$ after possibly reversing orientation. Every smooth point is a limit of z_ϵ's and w_ϵ's, smooth points are dense, and component boundaries lie in γ; so $\gamma = \partial\Omega = \partial\mathcal{U}$. Write $\Omega_\gamma = \Omega$, $\mathcal{U}_\gamma = \mathcal{U}$.

Step Ω_γ is simply connected. By Theorem 1.2 it suffices that $\Omega_\gamma^c = \gamma \cup \mathcal{U}_\gamma$ be connected. Suppose $\Omega_\gamma^c = F_1 \cup F_2$, closed, disjoint, non-empty; then $\mathcal{U}_\gamma = O_1 \cup O_2$, $O_i = \mathcal{U}_\gamma \cap F_i$ disjoint.

O_1 is open: small balls about $z \in O_1$ lie in $\mathcal{U}_\gamma \subset F_1 \cup F_2$, and if all met F_2 then $z \in F_2$; so one lies in O_1. Likewise O_2.

$O_1 \neq \emptyset$. Otherwise $F_1 \subset \gamma$ and $\mathcal{U}_\gamma \subset F_2$. Take $z \in F_1 \subset \gamma$. Since $\gamma = \partial\mathcal{U}_\gamma$, every ball at z meets $\mathcal{U}_\gamma \subset F_2$, so $z \in F_2$, a contradiction. Likewise $O_2 \neq \emptyset$.

This disconnects $\mathcal{U}_\gamma$, a contradiction. This completes the proof of the Jordan curve theorem for piecewise-smooth curves. ∎

2.1 Proof of a general form of Cauchy's theorem

Theorem 2.9

If f is holomorphic in an open set containing a simple closed piecewise-smooth curve γ and its interior, then $\int_\gamma f = 0$.

Let f be holomorphic on an open $\mathcal{O} \supset \gamma \cup \Omega_\gamma$. Build a closed $\Gamma \subset \Omega_\gamma$ so close to γ that $\int_\gamma f = \int_\Gamma f$, which is 0 by Theorem 2.3. Near smooth parts Γ is essentially γ_η (Lemma 2.4); near junctions it follows circular arcs.

Lemma 2.10

Let $\gamma : [0,1] \to \mathbb{C}$ be a simple smooth curve. For all sufficiently small $\epsilon > 0$ the circle C_ϵ of center $\gamma(0)$ and radius ϵ meets γ in exactly one point.

Proof. Assume $\gamma(0) = 0$. For $\epsilon < |\gamma(1)|$ the intermediate value theorem gives an intersection. If $|\gamma(t)| = \epsilon_j$ had two solutions for $\epsilon_j \to 0$, both would tend to 0 ($|\gamma(t)| \geq c(\tau) > 0$ for $t \geq \tau$, simplicity), and the mean value theorem for $h = |\gamma|^2$ gives $t_j \to 0^+$ with $h'(t_j) = 2\,\mathrm{Re}[\gamma'(t_j)\overline{\gamma(t_j)}] = 0$. But $\gamma(t) = \gamma'(0)t + t\varphi(t)$, $\gamma'(t) = \gamma'(0) + \psi(t)$, $\varphi, \psi \to 0$, so

$$\mathrm{Re}[\gamma'(t)\overline{\gamma(t)}] = |\gamma'(0)|^2 t + o(t),$$

which is $\neq 0$ for small $t > 0$ since $\gamma'(0) \neq 0$. This is a contradiction. ∎

Proof of Theorem 2.9. Choose ϵ with $U = \{d(z,\gamma) < \epsilon\} \subset \mathcal{O}$. Let $P_1, \ldots, P_n$ be the consecutive junction points, and $\delta < \epsilon/10$ so small that the circle C_j of radius δ about P_j meets γ in exactly two points (Lemma 2.10 for the two parts at P_j; the rest of γ is farther away). Exactly one of the two arcs, $\mathcal{C}_j$, has interior in Ω_γ: for small η, $\gamma_{\pm\eta}$ near P_j lie in different components (proof of Theorem 2.2) and both meet C_j. The disc D_j of radius 2δ about P_j lies in $U \subset \mathcal{O}$.

Let B_j be the end-point of $\mathcal{C}_j$ on the part γ_j from P_j to P_{j+1}, and A_{j+1} that of $\mathcal{C}_{j+1}$ on γ_j. Take a chain of discs $\mathcal{D} = \{D_0, \ldots, D_K\}$ in U covering γ, $D_k \cap D_{k+1} \neq \emptyset$, $D_0 = D_K$, including the D_j. By Lemma 2.4 a curve Γ_j in $\Omega_\gamma \cap \bigcup \mathcal{D}$ runs parallel to γ_j from near B_j on $\mathcal{C}_j$ to near A_{j+1} on $\mathcal{C}_{j+1}$ (replaced by a nearby polygonal line, as γ is only C^1). Joining the Γ_j along pieces of the $\mathcal{C}_j$ gives a closed piecewise-smooth $\Gamma \subset \Omega_\gamma$.

f has a primitive on each disc of $\mathcal{D}$ (Chapter 2, Theorem 2.1); as in Chapter 3, Theorem 5.1, the primitives on consecutive discs differ by constants and the endpoint contributions telescope, so $\int_\gamma f = \int_\Gamma f = 0$ (Theorem 2.3). ∎